LCD Board Power Budget - PRELIMINARY / Revision H

Team Number:	202
Project Name:	**Weather Gadget - LCD/button board**
Team Member Names:	**Zander Boward**
Version:	**2026-10-31; Revision H; provisional load inputs**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements

All Major Components	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Nano target (PROVISIONAL)	DM164150 / PIC18F57Q43	5 V design	1	50.000	50.000	mA
	LCD logic (PROVISIONAL)	SunFounder CN0293 / DS1	5 V logic	1	5.000	5.000	mA
	LCD backlight (PROVISIONAL)	Same CN0293 / DS1	3.3 V backlight	1	50.000	50.000	mA
	Schmitt buffer + transition allowance	SN74LVC1G17DBVR / U4	5 V	1	0.510	0.510	mA
	Indicators D2-D5 (one group)	RED / 0805; branches below	VIN / 5 V / 3.3 V	1	14.159	14.159	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.
Add additional power rails or change the power rail voltages if needed.

Unused power rail	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
						0.000	
						0.000	
						0.000	
						0.000	
						0.000	
					Not applicable - no devices on this rail		

c1. Not applicable

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Nano target (PROVISIONAL)	DM164150 / PIC18F57Q43	5 V	1	50.000	50.000	mA
	CN0293 LCD logic (PROVISIONAL)	SunFounder CN0293 / DS1	5 V	1	5.000	5.000	mA
	U4, indicators and bias networks	U4 + support subtotal below	5 V	1	10.811	10.811	mA
	Complete 3.3 V downstream load	Unmargined 3.3 V subtotal	5 V input path	1	53.542	53.542	mA
	LD1117 ground current	LD1117V33	5 V input path	1	10.000	10.000	mA
					Subtotal	129.353	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	161.691	mA
c2. Regulator or Source Choice	5 V regulator (nameplate rating)	L7805CV	9-12 V input design	1	1500	1500.000	mA
					Total Remaining Current Available on +5V Rail	1338.309	mA

Unused power rail	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
						0.000	
						0.000	
						0.000	
					Not applicable - no devices on this rail		

c3. Not applicable

+3.3V Power Rail	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	CN0293 backlight (PROVISIONAL)	SunFounder CN0293 / DS1	3.3 V	1	50.000	50.000	mA
	3.3 V indicator D4	R15 = 1k, +/-5%	3.3 V	1	3.542	3.542	mA
						0.000	
						0.000	
					Subtotal	53.542	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	66.928	mA
c4. Regulator or Source Choice	3.3 V regulator (nameplate rating)	LD1117V33	5 V nominal input	1	800	800.000	mA
					Total Remaining Current Available on 3.3V Rail	733.072	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each rail above is not negative.

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need multiple power sources, list each separately below and indicate which regulators will be connected to each supply. Confirm that the Total Remaining Current Available on each power source below is not negative.

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	9 V wall adapter	Triad WSU090-1300	100-240 VAC	9 VDC	1300	1300.000	mA
Power Rails Connected to External Power Source 1	5 V stage demand +25%	L7805CV	9 V nominal input	1	161.691	161.691	mA
	L7805 ground allowance +25%	L7805CV	Input path	1	12.500	12.500	mA
	Input indicator D2 +25%	R13 = 4.7k, +/-1%	12 V current ceiling	1	3.224	3.224	mA
			Total Remaining Current Available on External Power Source 1			1122.585	mA

External Power Source 2	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 2 Selection	Not applicable - only one external source						
Power Rails Connected to External Power Source 2							

E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the battery by indicating the capacity in mAh.

Component Name	Part Number	Supply Voltage Range	Capacity (mAh)	Required By Regulators
Not applicable - wall powered				

Notes

PRELIMINARY: F9=50, F10=5 and F11=50 mA are editable planning allowances, NOT established absolute maximum currents. Nano and CN0293 limits remain unverified.

CN0293 is one display: logic uses 5 V and backlight uses 3.3 V. Its two current entries represent separate supplies, not two modules. Newhaven current figures do not apply.

Support ceilings use 5.25 V, 3.365 V and 12 V; resistor -5% except R13 -1%; RV1 -10%; LED Vf=0. These deliberately conservative branch ceilings are not measured currents.

R5-R9 allowance assumes full rail on every pull-down. R/W is normally LOW and DB0-DB3 unused, so this is a conservative provision. RX LOW, E HIGH, button held and D5 ON are budgeted.

U4 uses 10 uA static ICC plus a provisional 500 uA transition allowance. Validate the slow RC input current. Nano allocation excludes external D5 and bias currents counted separately.

The 5 V path includes the entire UNMARGINED 3.3 V load plus LD1117 ground current. Apply 25% only once per physical path. Input adds L7805 ground allowance and D2.

F1=250 mA input; F2=500 mA 5 V; F3=500 mA backlight. Include diode, fuse, link and contact voltage losses. F2 reduces U3 headroom. Fuse inrush, derating and clearing remain unqualified.

Remaining-current rows are nameplate rating minus planning demand, not guaranteed usable spare capacity. Verify voltage and cooling; U2 dissipates about 1.025 W at 12 V in this illustration.

Board only: J1-J3 export no power; no J5 external load is included. USB debugger is powered separately. Battery life is not applicable. Indicator group in A is allocated by branch in B/D.

Sources: SunFounder CN0293 product page; ST L78 Table 10; ST LD1117 Table 6; TI SN74LVC1G17; Triad WSU090-1300; current PCB BOM resistor tolerances. No bench measurements claimed.

Supporting assumptions and calculations (mA unless stated)

Leakage / I/O current reserve (mA)	0.1		mA
L7805 ground current allowance (mA)	10		mA
LD1117 ground current allowance (mA)	10		mA
U4 static ICC ceiling (mA)	0.01		mA
U4 transition current allowance (mA)	0.5		mA

Support load	Resistance (ohm)	Tolerance	#	Voltage (V)	Current (mA)	Unit
D2 / R13 input indicator	4700	1%	1	12.000	2.579	mA
D3 / R14 5 V indicator	2200	5%	1	5.250	2.512	mA
D4 / R15 3.3 V indicator	1000	5%	1	3.365	3.542	mA
D5 / R16 GPIO indicator	1000	5%	1	5.250	5.526	mA

R1-R3 RX pullups	47000	5%	3	5.250	0.353	mA
R10 LCD E pull-down	47000	5%	1	5.250	0.118	mA
RV1 contrast trimmer	50000	10%	1	5.250	0.117	mA
R11 button pullup	10000	5%	1	5.250	0.553	mA
R5 LCD R/W pull-down	10000	5%	1	5.250	0.553	mA
R6-R9 LCD DB0-DB3 bias	47000	5%	4	5.250	0.470	mA
			Direct 5 V support loads + leakage (excludes U4)		10.301	mA